

AI Powered Smart Interview Assessment: A Multi-Modal Framework for Automated Candidate Evaluation

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Abstract—Traditional candidate recruitment and screening processes rely heavily on manual evaluation, making them time-consuming, difficult to scale, and vulnerable to subjective human bias. This paper presents a multi-modal framework for the *AI Powered Smart Interview Assessment* system, designed to automate candidate evaluations across resume screening, dynamic AI interviewing, and practical skill assessment. By integrating Natural Language Processing (NLP) for answer relevancy, speech analytics for delivery confidence, and code execution grading, the system generates an objective, unified scorecard. Experimental execution on a functional Python prototype demonstrates effective score differentiation based on candidate answer relevancy and response timing metrics.

Index Terms—Multi-Modal AI, Natural Language Processing, Automated Candidate Screening, Speech Analytics, Resume Matching, Score Fusion Engine.

I. INTRODUCTION

Modern talent acquisition faces severe scalability challenges when managing high volumes of applicants[cite: 2]. Traditional Applicant Tracking Systems (ATS) rely primarily on rigid keyword matching, frequently missing qualified candidates who express relevant skills using alternative terminology[cite: 2]. Furthermore, subsequent interview stages lack standardized, objective measurement criteria across candidate responses, communication delivery, and technical skill execution[cite: 2].

Manual review cycles not only extend the time-to-hire window but also introduce implicit cognitive biases[cite: 2]. To address these structural limitations, this project proposes an end-to-end multi-modal assessment architecture[cite: 2]. By combining semantic resume screening, dynamic AI-driven questioning, speech delivery analysis, and automated coding evaluation into a unified processing pipeline, the system minimizes human intervention while providing transparent, data-driven decision support for hiring teams[cite: 2].

II. LITERATURE SURVEY & RELATED WORK

Automated hiring systems have evolved from basic rule-based parsers into complex multi-modal intelligence engines[cite: 2]. Existing approaches in literature can be categorized into four domain streams:

A. Keyword Matching vs. Semantic ATS

Traditional ATS tools parse resumes using regex patterns or exact string matching[cite: 2]. Recent studies demonstrate that Transformer-based language models (e.g., BERT, RoBERTa) significantly outperform keyword matching by measuring vector similarity in contextual embedding spaces[cite: 2].

B. Speech Recognition and Acoustic Analysis

Standalone Automatic Speech Recognition (ASR) engines convert spoken interview answers into text[cite: 2]. However, conventional ASR pipelines ignore non-verbal audio parameters such as speech latency, pitch variation, and filler word frequency, which serve as key indicators of candidate confidence[cite: 2].

C. Automated Code Evaluation

Platforms like HackerRank and LeetCode evaluate technical competencies by running candidate submissions against predefined test cases[cite: 2]. While effective for isolated coding tasks, these platforms operate independently from soft-skill and interview communication evaluations[cite: 2].

D. Research Gap

Current industry tools operate in silos[cite: 2]. Resume screening, spoken interview evaluation, and technical coding platforms rarely merge into a unified pipeline[cite: 2]. This work bridges this gap by proposing a single, explainable multi-modal scoring framework[cite: 2].

III. PROPOSED METHODOLOGY

The system processes candidate data through five distinct, interconnected modules[cite: 2]:

A. Intelligent Resume Screening

Candidate resumes (PDF/DOCX) and Job Descriptions (JDs) are parsed into plain text[cite: 2]. Semantic vectors are generated using sentence-transformer models to calculate cosine similarity scores[cite: 2]:

$$\text{Similarity Score} = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} \quad (1)$$

Candidates meeting the minimum threshold pass directly to the AI interview phase[cite: 2].

B. Multi-Modal Feature Extraction Engine

During the interview, three data streams are recorded and analyzed simultaneously[cite: 2]:

- **Text Stream (NLP):** Evaluates answer relevance against ideal domain responses using BERT embeddings[cite: 2].
- **Audio Stream (Speech Analytics):** Uses OpenAI Whisper for transcription while extracting timing metadata (response duration, silence latency)[cite: 2].
- **Visual Stream (Computer Vision):** Captures frame streams via OpenCV to track head orientation and facial sentiment indicators[cite: 2].

C. Technical Skill Validation

Candidates complete practical coding challenges within an integrated sandbox[cite: 2]. The evaluation engine executes code against functional unit tests to grade correctness, execution time, and memory efficiency[cite: 2].

D. Score Fusion Engine

The sub-scores are aggregated into a single candidate evaluation index (S_{Overall}) using weighted parameter matrices[cite: 2]:

$$S_{\text{Overall}} = w_1 S_{\text{Resume}} + w_2 S_{\text{Text}} + w_3 S_{\text{Speech}} + w_4 S_{\text{Code}} \quad (2)$$

Where $w_1 + w_2 + w_3 + w_4 = 1.0$. In our baseline setup, weights are configured as 60% content relevance (S_{Text}) and 40% communication delivery (S_{Speech})[cite: 2].

IV. IMPLEMENTATION AND RESULTS

A. Prototype Implementation

A core evaluation prototype was built using Python 3, Pandas, and NumPy[cite: 2]. The script processes transcript responses, extracts technical domain terms, and measures delivery timing windows[cite: 2].

B. Experimental Execution Results

The evaluation engine was tested against multiple test responses. The output metrics generated by the pipeline are summarized in Table I.

TABLE I
CANDIDATE ASSESSMENT EXECUTION RESULTS

Candidate ID	Keywords	Content	Delivery	Overall	Status
CAND_01	4	100.0	90.0	96.0	Recommended
CAND_02	0	0.0	65.0	26.0	Rejected
CAND_03	3	100.0	90.0	96.0	Recommended

C. Discussion

The prototype successfully differentiates between high-relevance and low-relevance candidate responses[cite: 2]. Candidates who incorporate technical concepts within recommended timing windows earn scores above 90.0, whereas superficial responses receive significantly lower ratings[cite: 2].

V. CONCLUSION AND FUTURE WORK

This paper presents a multi-modal assessment framework that automates candidate evaluations across resume screening, spoken interviews, and coding tests[cite: 2]. The prototype demonstrates that combining text relevance scoring with delivery metrics provides an objective alternative to manual initial screening[cite: 2].

Future development will focus on:

- 1) Integrating real-time OpenAI Whisper ASR models for live voice-to-text processing[cite: 2].
- 2) Incorporating deep learning facial emotion tracking via OpenCV[cite: 2].
- 3) Building a cloud-hosted web interface for recruiter dashboard analytics[cite: 2].

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